



git



COMP1117

Tutorial 4: GitHub, Version control



Introduction

This tutorial offers a concise overview of GitHub and Version Control for Beginners. You will know the basic operation of setting up your own Git repository, saving changes to your repository, commit, pushing and pulling after this tutorial.



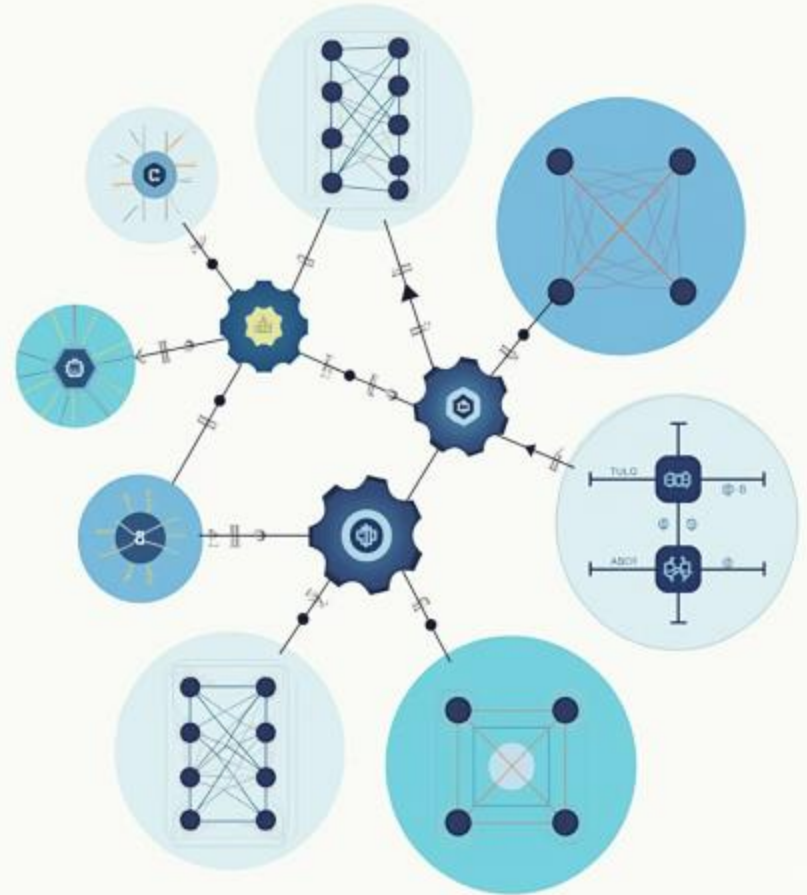
What you are expected to learn?

- What version control is
- Why version control is important
- Difference between **Git** and **GitHub**
- How to create and use a repository
- How to **clone**, **commit**, **push**, and **pull**

This tutorial is intended for beginners and you do not need any prior knowledge

01

Version Control





What is version control ?

Version control functions as a tracking mechanism that monitors document modifications throughout a project's lifecycle.

It assists you with

- **History Preservation:** Maintains a complete log of every alteration made.
- **Rollback Capabilities:** Allows you to effortlessly revert to previous iterations.
- **Collaborative Safety:** Enables multiple contributors to work simultaneously without risking data overwrites.
- **Broad Applications:** Ideal for managing software code, written documents, and complex team projects.

What is version control ?

Instead of saving files as messy naming conventions like:

- project_final.docx
- project_final2.docx
- project_final_reallyfinal.docx

Version control keeps everything organized automatically.



Why Do We Need Version Control?

Why Version Control Matters, primary advantages include:

- **Data Security:** Prevent accidental data loss and protect your progress.
- **Modification Tracking:** Monitor exact updates and the timestamps of when they occurred.
- **Accountability:** Identify which contributor made specific edits.
- **Seamless Collaboration:** Partner with colleagues simultaneously while safeguarding against data overwrites.
- **Instant Recovery:** Roll back to functional previous states whenever errors occur.

02

Git VS GitHub





Git VS GitHub

Git and GitHub are NOT the same !

Git: A local utility installed directly on your machine designed to monitor file modifications.

GitHub: An online platform and web service dedicated to hosting Git repositories remotely.

Git Function: Tracks changes.

GitHub Function: Assists in share, store and team collaboration for your Git projects.

Git tracks file changes and manages versions while Github stores your repositories online for sharing.



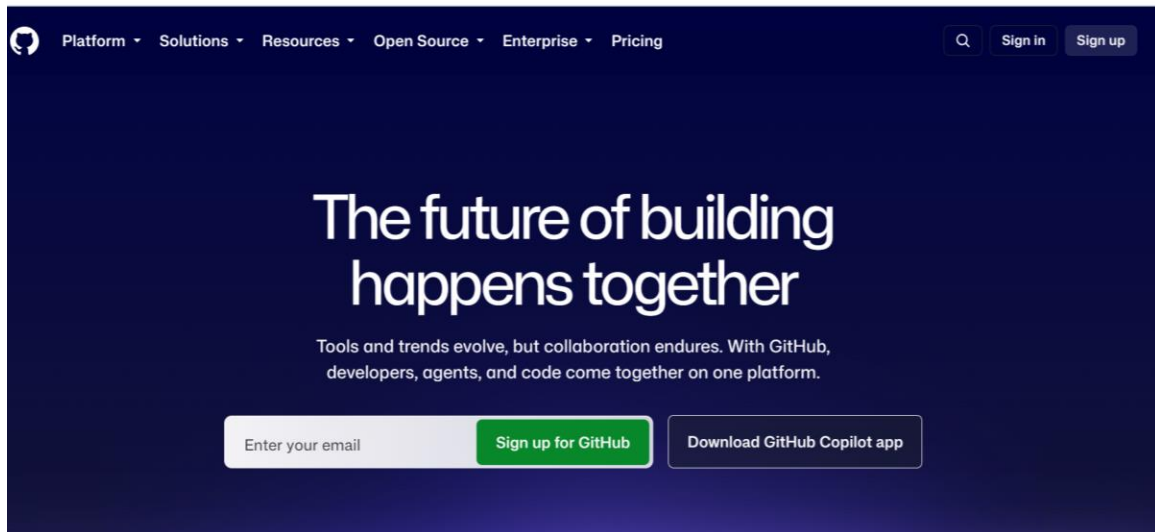
Important Terms

- **Repository (Repo):** Project folder tracked by Git
- **Commit:** A saved snapshot of changes
- **Clone:** Copy a repository to your computer
- **Pull:** An action to download changes from GitHub
- **Push:** An action to upload changes to GitHub

These are official terms !

Create Your GitHub Account

1. Visit <https://github.com>
2. Click **Sign up**
3. Enter:
 - username
 - email
 - password
4. Verify your email
5. Log in to GitHub

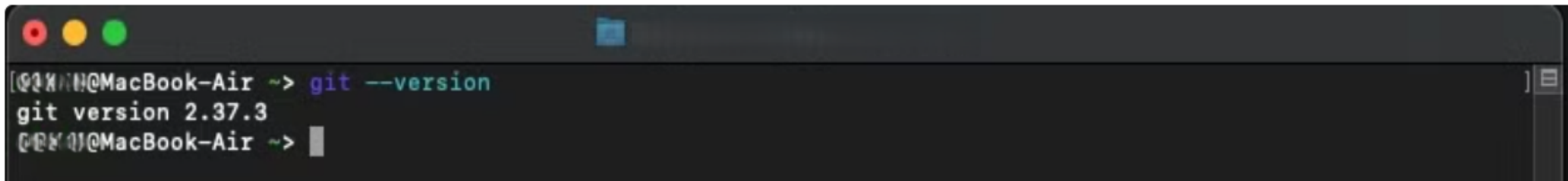


Install Git

How to install Git on your computer?

- Download Git from: <https://git-scm.com>
- Install using the default settings
- (Optional, but suggested) After installation, open:

Terminal (Mac/Linux) by pressing **Command + Spacebar**, type **Terminal**, and press **Return**, type **git --version** to check if the installation is successful

A screenshot of a macOS Terminal window. The title bar shows three colored window control buttons (red, yellow, green) and a blue folder icon. The terminal text shows the command 'git --version' being executed, resulting in the output 'git version 2.37.3'. The prompt is '[BM] MacBook-Air ~>'.

```
[BM] MacBook-Air ~> git --version
git version 2.37.3
[BM] MacBook-Air ~> 
```

Install Git

Or **by pressing Win + R**, then type cmd and press **Enter** (Windows) to open the command prompt.

In the command prompt, type git --version to check if the installation is successful

```
C:\Users\user>git --version  
git version 2.55.0.windows.5
```

Configure Git

Set up your Git identity by running the following command once in Terminal (Mac) or Command Prompt (Windows):

```
1 git config --global user.name "Your Name"
2 git config --global user.email "you@example.com"
```

For Windows, make sure Git is added to Windows PATH

Inspect your settings by entering below command in Terminal(Mac) or Command Prompt (Windows):

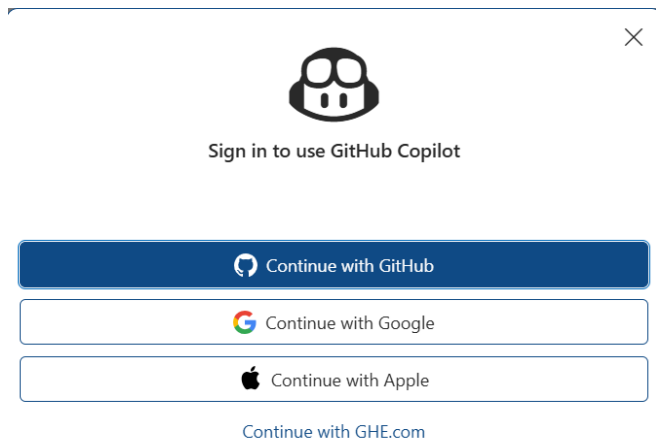
```
1 git config --list
```

Suggested to use the same email as your GitHub account during configuration if possible.

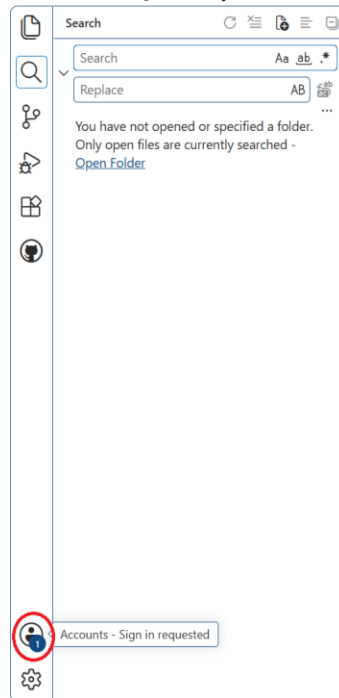
Sign in to GitHub inside VS Code

Sign in to GitHub inside VS Code by

- Click the Accounts icon located at the very bottom-left corner in VS Code
- Select Sign in with GitHub (or Sign in to GitHub or Sign in to use GitHub Copilot) from the pop-up menu.
- Click Continue with GitHub to sign in with your GitHub account



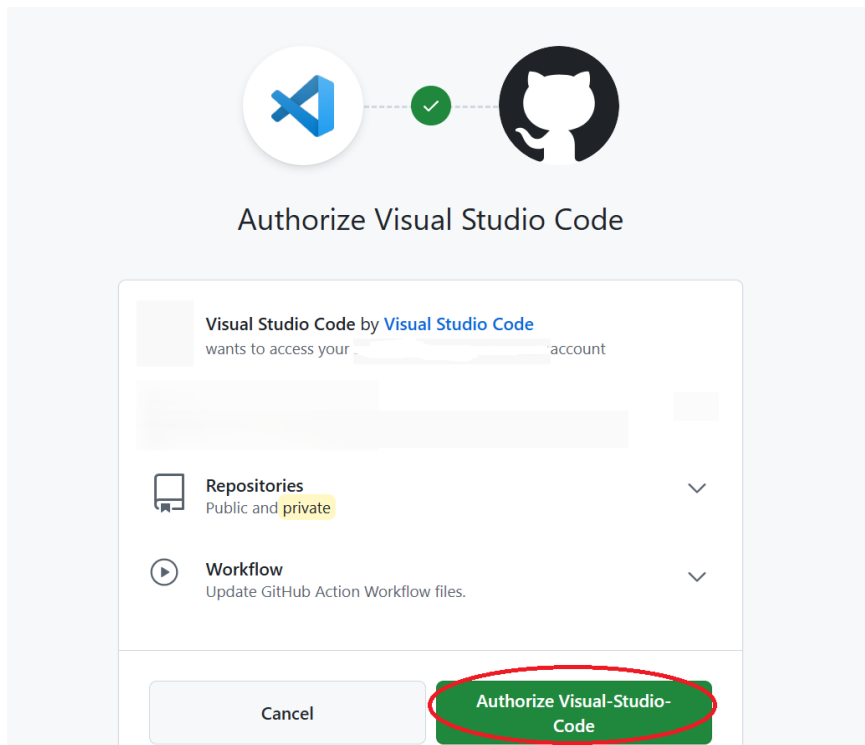
By continuing, you agree to GitHub's [Terms](#) and [Privacy Statement](#). GitHub Copilot may show [public code](#) suggestions and use your data to improve the product. You can change these [settings](#) anytime.



Sign in to GitHub inside VS Code

Sign in to GitHub inside VS Code by

- You may reach a page for authorization, please click authorize VS Code

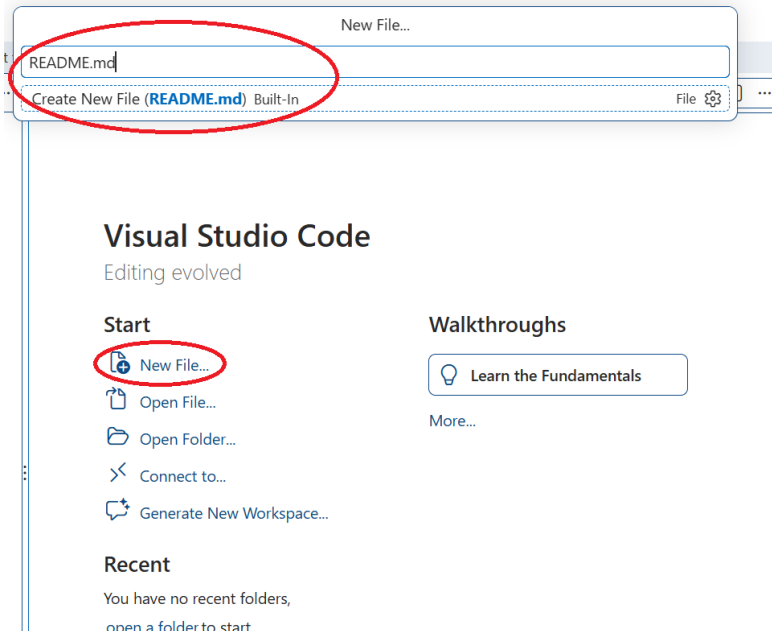


Create a new project and Git repository in VS Code

Step 1: Create a folder on your computer, for example with the name "my-project"

Step 2: Open VS Code. Then go to: File > Open Folder, select the newly created folder called "my-project"

Step 3: Inside VS Code, create a new file called "README.md"

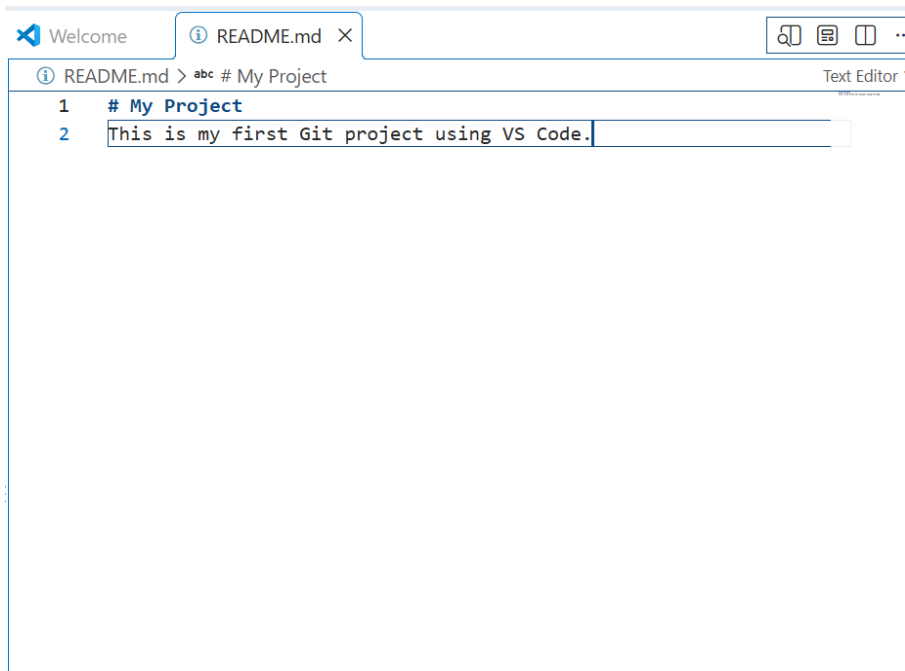


Create a new project and Git repository in VS Code

Step 4: Add below text and save the file

My Project

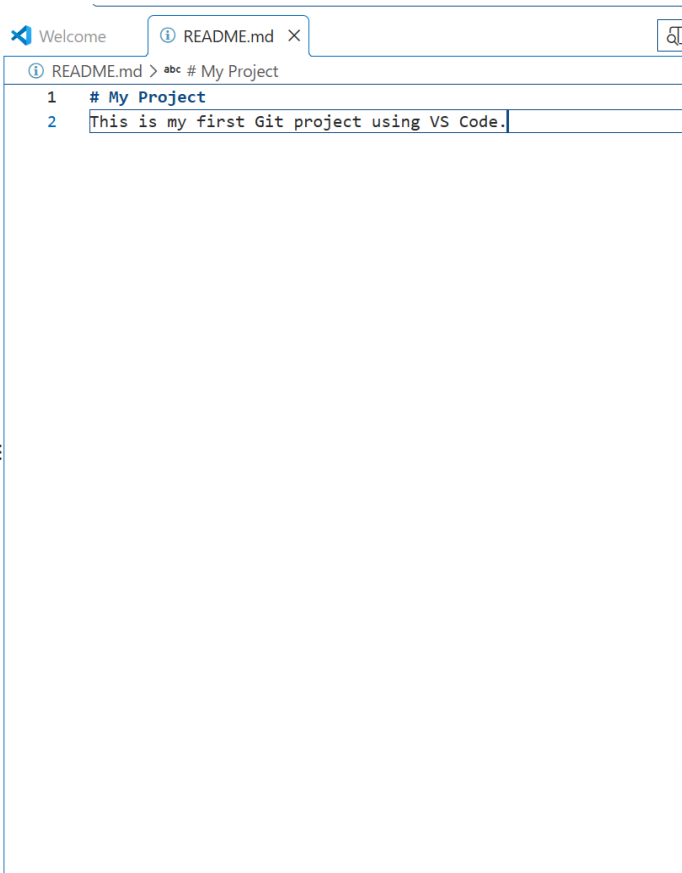
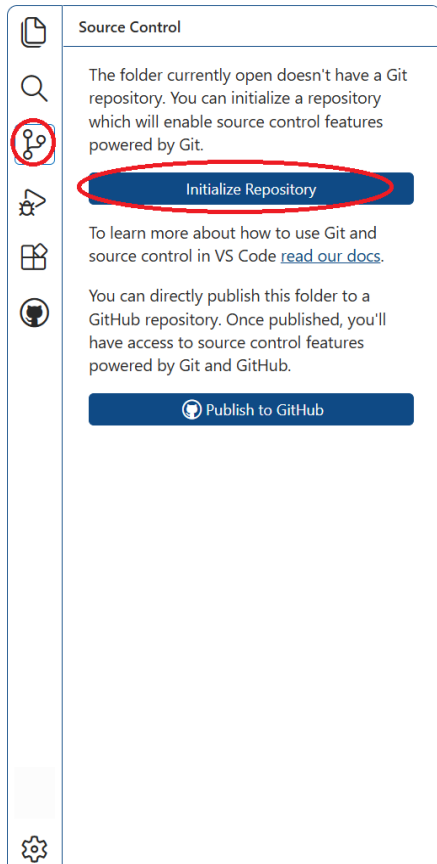
This is my first Git project using VS Code.



Initialize Git repository

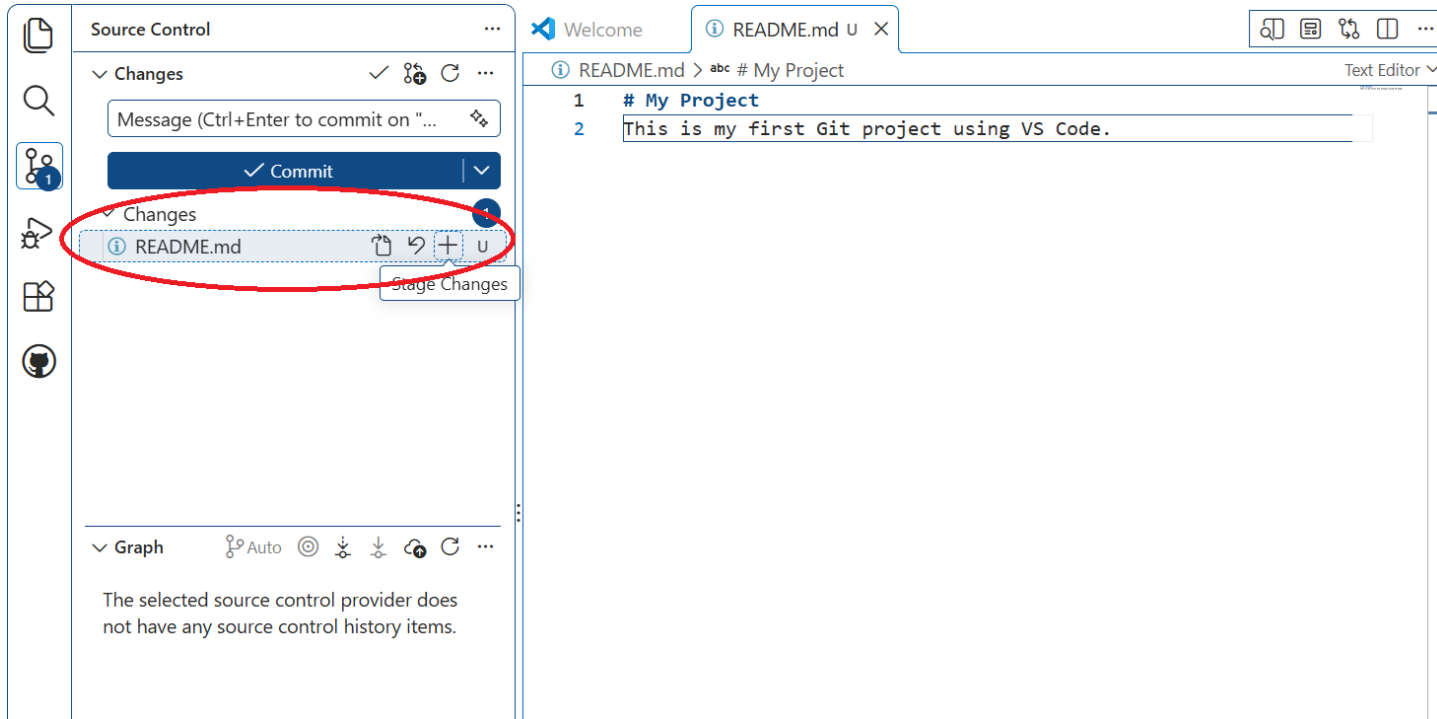
Step 1: Click the Source Control icon on the left sidebar. If the folder is not already a Git repository, you will see a button: Initialize Repository, click that button, VS Code has now created a local Git repository in that folder.

Note: In some case, you may be in Restricted Mode, please click "Trust" in order to allow VS Code to do the initialization.



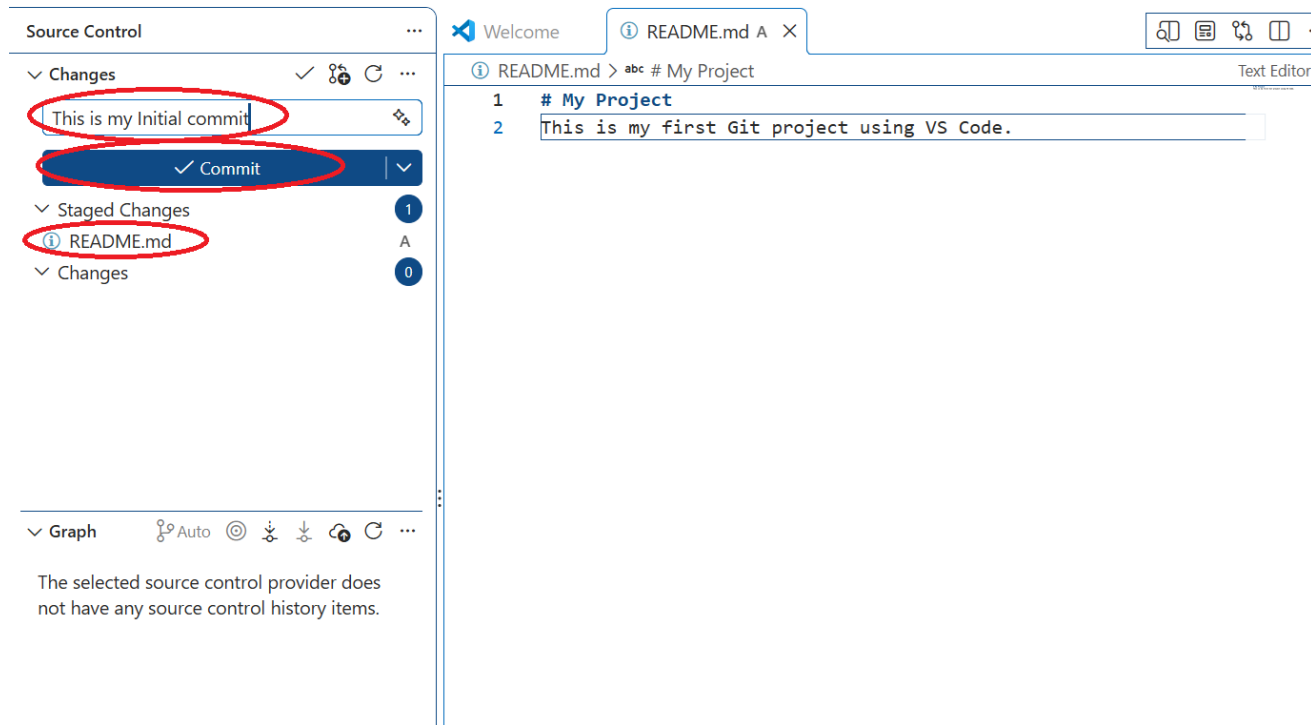
Commit changes in VS Code

Go to Source Control again, You should see your changed file, Hover over the file and click the + icon. This will moves the file into the staged area.



Commit changes in VS Code

Write a commit message, i.e. At the top of the Source Control panel, enter a message like: "This is my Initial commit", click the checkmark button, you have now made your first commit.



Publish your project to GitHub from VS Code

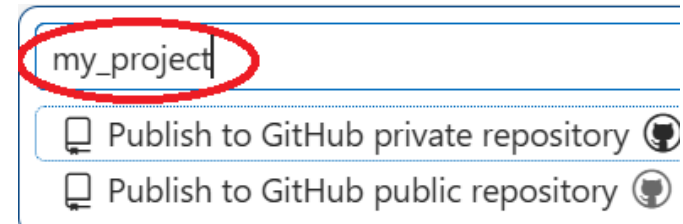
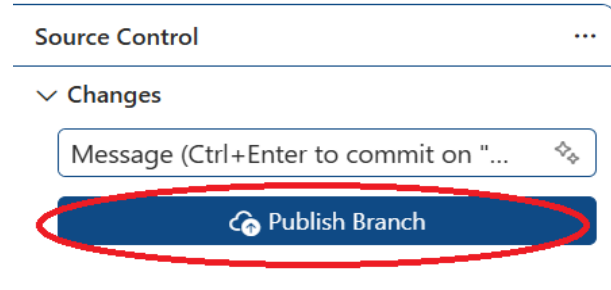
Click the "Publish Branch" button. VS Code may ask:

- Repository name
- Public or private repository

"my-project" could be the Repository name. Choose either:

- **Private**: only you and the one you invited can see it OR
- **Public**: All person can see it

After that, VS Code will upload your code and create the GitHub repository

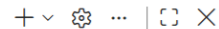


Welcome

① README.md X



Chat



① README.md > abc # My Project

Text Editor

1 # My Project

2 This is my first Git project using VS Code.

Sessions



Tip: Use [/create-agent](#) to scaffold a custom agent for your workflow.

+ ① README.md

Describe what to build

+ Agent A... Balance ↕ ↑

① Publish to GitHub: Uploading files

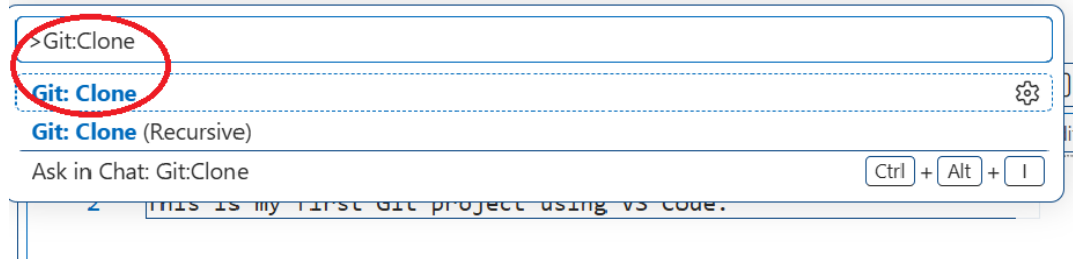
Clone the Repository

Cloning means downloading a GitHub repository to your computer.

Clone in VS Code:

In VS Code, open the Command Palette (Ctrl + Shift + P), search for Git: Clone

Then, choose the repo project which you want to clone. In our case, it is "my-project". Choose a folder where you want to save the project. VS Code will download the repository.



After cloning, please click open, then you can work on the project.

Clone the Repository

What clone is doing:

- downloads the repository
- creates a local copy
- links it to GitHub

What Is Inside the Repository?

After cloning, your folder may contain:

- **README.md** which is a project description
- a hidden **.git** folder



.git

9/9/2026 1:59



README.md

9/9/2026 1:59

Please note:

- Git uses the hidden .git folder to track history
- Do **not** make any changes to the .git folder manually

Make changes, commit, and push

Edit a File

Change a file in the project.

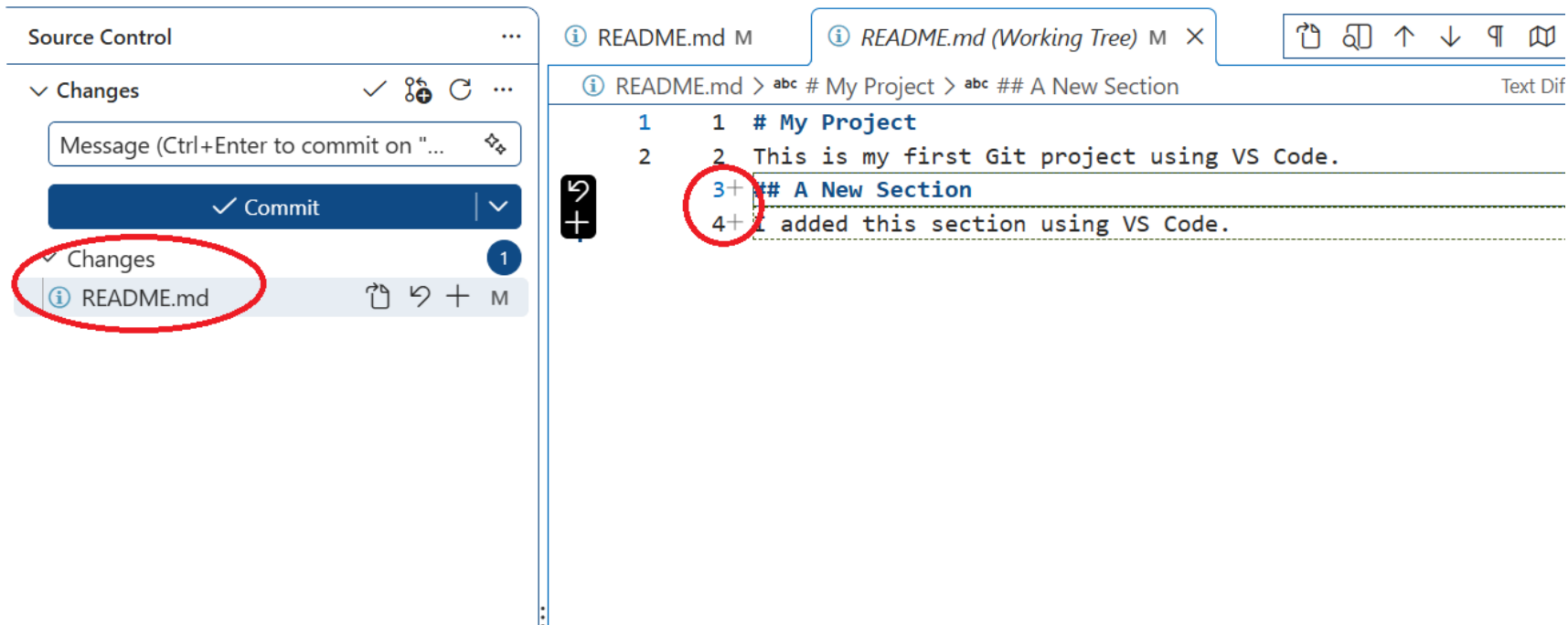
For example, edit README.md by adding the following and save the file afterwards:

```
## A New Section
```

```
I added this section using VS Code.
```

Go to the Source Control panel. You should see the modified file "README.md" listed under **Changes**. Click the file to view what is being changed.

Make changes, commit, and push



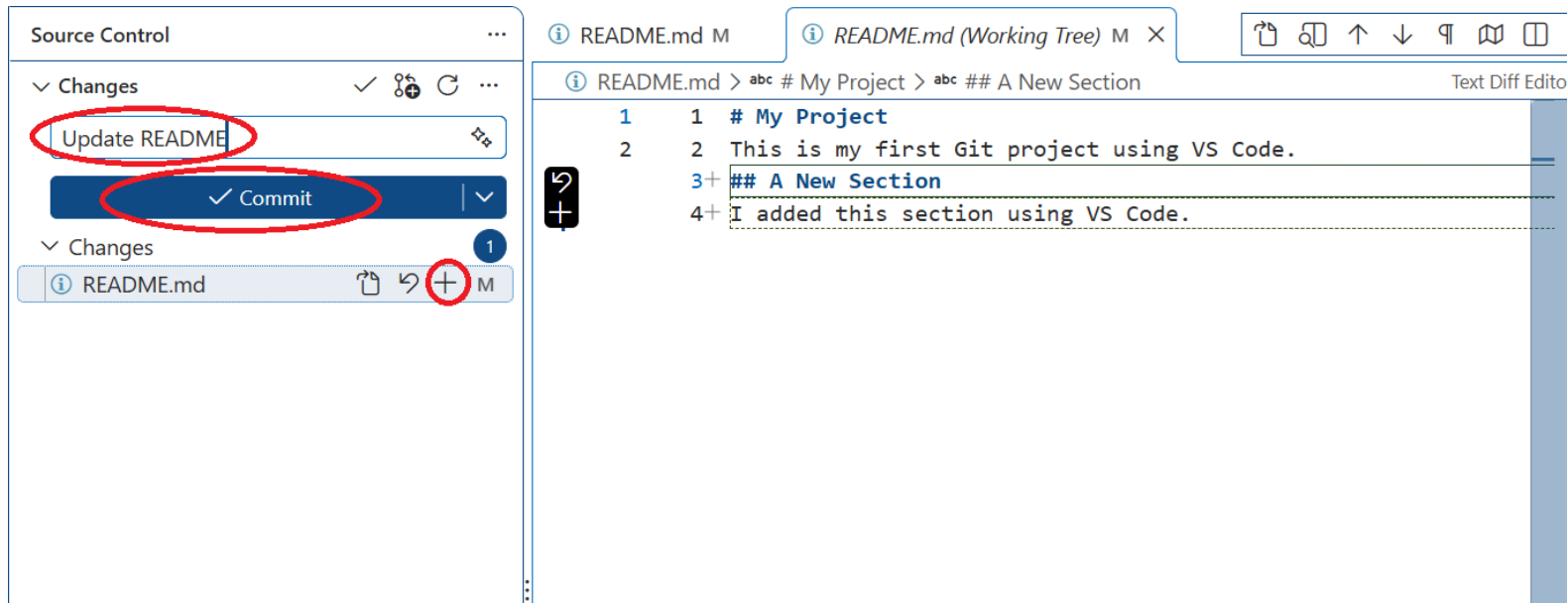
The screenshot displays the Visual Studio Code Source Control interface. On the left, the 'Source Control' sidebar is open, showing a list of changes under the 'Changes' section. The file 'README.md' is selected and circled in red. Above the list, there is a text input field for a commit message, a 'Commit' button, and a 'Changes' button, which is also circled in red. The main editor area shows a diff view for 'README.md'. The diff view has two tabs: 'README.md M' and 'README.md (Working Tree) M X'. The active tab shows the following content:

```
1 1 # My Project
2 2 This is my first Git project using VS Code.
3+3 # A New Section
4+4 I added this section using VS Code.
```

The lines 3 and 4, which represent the new additions, are circled in red. The diff view also includes a 'Text Dif' button in the top right corner.

Make changes, commit, and push

Stage the change by clicking the + icon next to the file.
Then commit the change by writing a commit message and click the checkmark to commit.

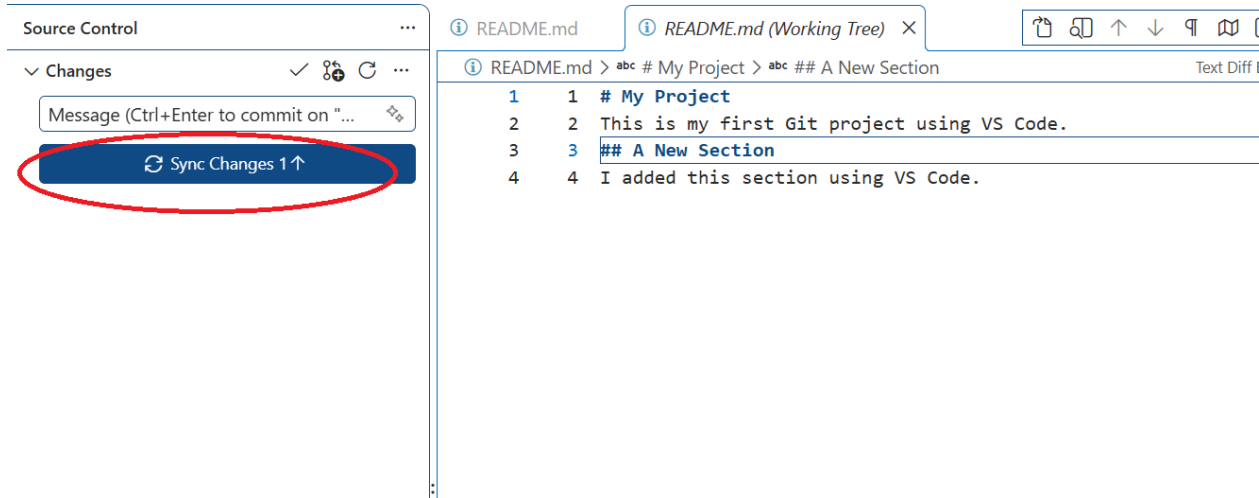


Make changes, commit, and push

What a commit does:

- saves a snapshot of your work
- creates a history entry
- adds a message to explain the change(s)

After committing, you need to push the changes to GitHub. You can just click Sync Changes in VS Code, Your commit is now uploaded to GitHub.



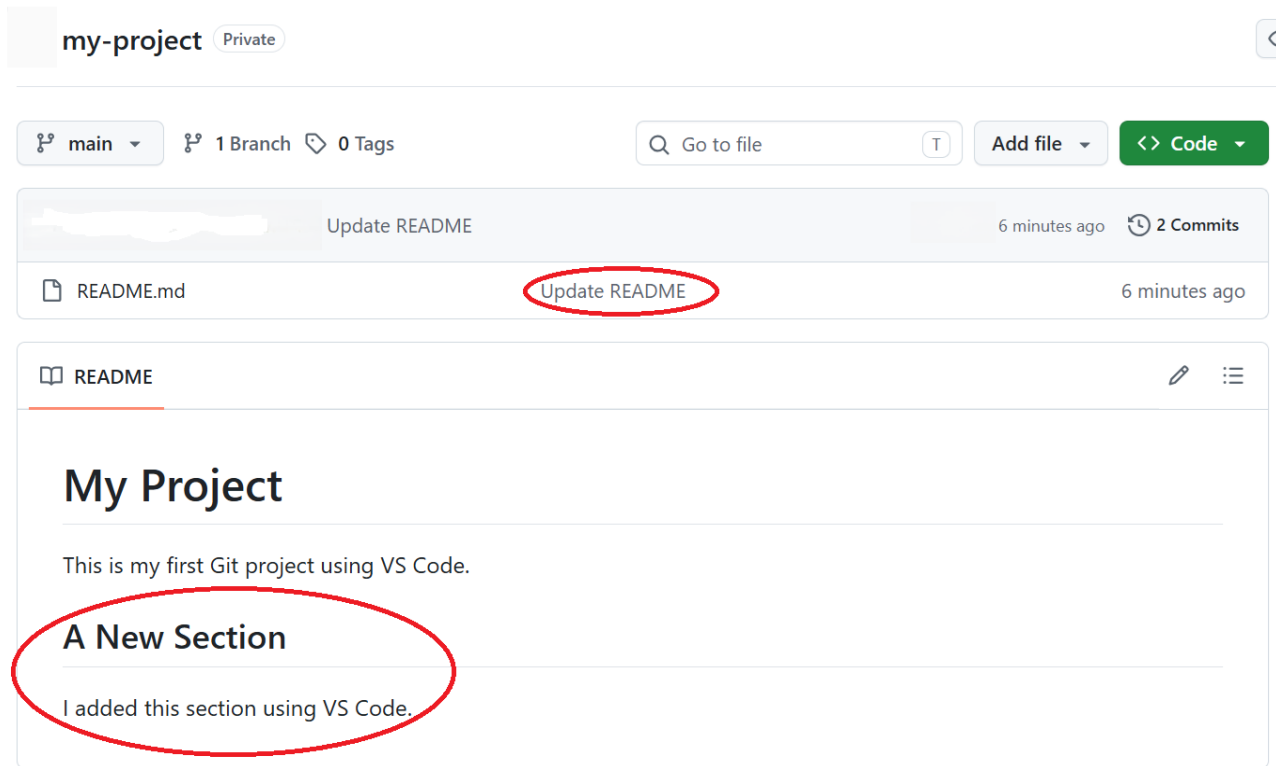
Push the Changes to GitHub

What push does:

- sends your local commits to GitHub
- updates the online repository

Next step:

- refresh the GitHub page
- confirm that your README file is updated



The screenshot shows the GitHub interface for a repository named 'my-project' (Private). At the top, it indicates 'main' branch, '1 Branch', and '0 Tags'. A search bar and 'Add file' button are visible. Below this, a commit history table shows a recent commit titled 'Update README' by a user, made '6 minutes ago' with '2 Commits'. The file 'README.md' is listed with an 'Update README' link circled in red. Below the commit list, the README content is displayed, showing a title 'My Project', a description 'This is my first Git project using VS Code.', and a new section titled 'A New Section' (circled in red) with the text 'I added this section using VS Code.'.

my-project Private

main 1 Branch 0 Tags

Go to file T Add file <> Code

Update README	6 minutes ago	2 Commits
README.md	Update README	6 minutes ago

README

My Project

This is my first Git project using VS Code.

A New Section

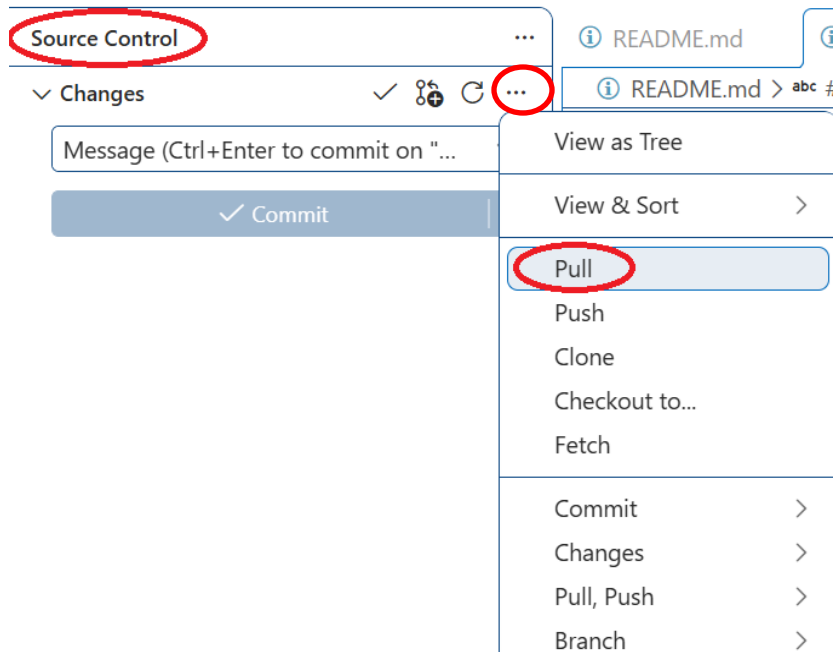
I added this section using VS Code.

Pull changes from GitHub

If you have changed something directly on GitHub, or someone else updated the repository, or you work from more than one computer, you need to do pulling. It means downloading the latest changes from GitHub to your computer.

In VS Code, you can do this by clicking the ... menu in Source Control. Choose: Pull.

VS Code will download and merge the latest changes.





Good practices for Beginners

When working in team projects, it is good to

- **Sync your local repository (Pull)** before kicking off any new tasks.
- **Save your progress frequently (Commit)** with descriptive and meaningful commit messages.
- **Maintain accurate documentation** by keeping the README file updated.
- **Inspect your modifications** thoroughly before pushing them to the remote server.



Common Git work flow in VS Code

If you work with others using Git, a common work flow may be the following:

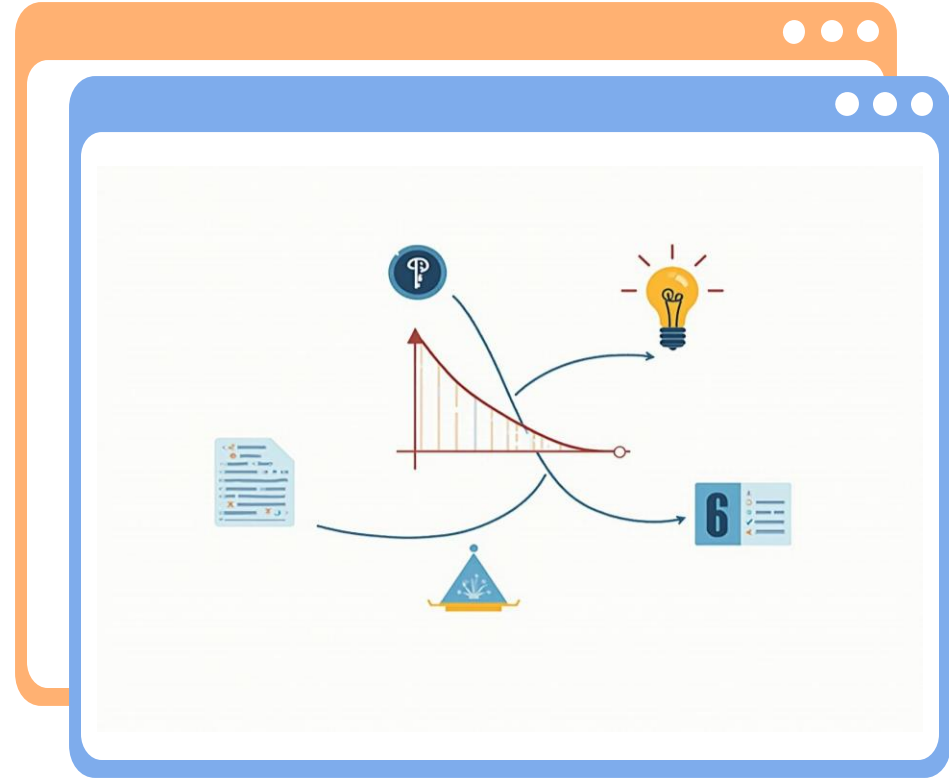
- Open your project in VS Code.
- Pull changes from GitHub
- Edit files.
- Save files.
- Stage changes.
- Write a commit message and commit.
- Push to GitHub.

Today

You have learnt:

- what version control is
- the difference between Git and GitHub
- how to create, clone, commit, push, pull in VS Code

You will learn more on GitHub in the topic 'Advanced GitHub'



References

A. Wanjala, "How to Check and Update Your Git Version on Linux, Mac, and Windows," *MakeUseOf*, Sep. 15, 2022. [Online].

Available: <https://www.makeuseof.com/git-update-version-linux-mac-windows/>

"Git," *Git-scm.com*. [Online]. Available: <https://git-scm.com/>. [Accessed: Sep. 26, 2026].

GitHub, Inc., *GitHub*. [Online]. Available: <https://github.com/>. [Accessed: Sep. 26, 2026].

Microsoft, *Visual Studio Code*. [Online]. Available: <https://code.visualstudio.com/>. [Accessed: Sep. 26, 2026].

The slide is framed by a thick orange border. At the top, there is a dark blue header bar. On the left side of the header bar, there are three small circles in orange, white, and teal. The word "THANKS!" is centered in a large, dark blue, sans-serif font.

THANKS!

CREDITS: This presentation template was created by [Slidesgo](#), including icons, infographics & images by [Freepik](#) and illustrations